

Model Curriculum

Helper- Wet Operations (Multiskilled)

SECTOR: LEATHER
SUB-SECTOR: FINISHED LEATHER
OCCUPATION: WET OPERATIONS (ASSISTANCE)
REF ID: LSS/Q0901, V1.0
NSQF LEVEL: 2

Certificate

CURRICULUM COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the

LEATHER SECTOR SKILL COUNCIL

for the

MODEL CURRICULUM

Complying to National Occupational Standards of

Job Role/ Qualification Pack: **'Helper-Wet Operations (Finished Leather)'** QP No. **'LSS/Q0901 NSQF Level 2'**

Date of Issuance: April 9th, 2016
Valid up to*: April 10th, 2018

**Valid up to the next review date of the Qualification Pack*


Authorised Signatory
(Leather Sector Skill Council)

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Helper- Wet Operations (Finished Leather)

CURRICULUM / SYLLABUS

This program is aimed at training candidates for the job of a “Helper- Wet Operations (Finished Leather)”, in the “Leather” Sector/Industry and aims at building the following key competencies amongst the learner

Program Name	Helper - Wet Operations (Finished Leather)		
Qualification Pack Name & Reference ID.	Helper - Wet Operations (Finished Leather) LSS/Q0901, V 1.0		
Version No.	1.0	Version Update Date	27-09 – 2016
Pre-requisites to Training	Class V		
Training Outcomes	After completing this programme, participants will be able to: • Gain knowledge on overview of leather industry and generic skills: General instructions to importance of leather industry, general hazards in leather industry, basic health safety and hygiene precautions, generic skills to be followed in leather industry. • Identify and use basic tools, equipment & materials: Effectively identify, select & use the specified tools, materials and equipment relevant to wet operations. • Learn and assist various wet operations in finished leather: Assist in preservation, soaking, liming, re-liming, fleshing, de-liming, tanning to manufacture finished leather. • Handle work area, tools and machines appropriately: Ensure correct handling of material, tools and machines, maintain clean and hazard free work area, proper usage of cleaning equipment, maintain correct usage of workplace tools and equipment. • Become well versed with environment, health, safety & security at work: Well versed with health and safety measures in terms of personal safety and equipment safety relevant to wet operations. • Follow industrial regularity and organizational requirements: Carry out work in accordance with organizational guidelines and procedures, identify and report any possible deviation.		

This course encompasses 5 out of 5 National Occupational Standards (NOS) of “Helper-Wet Operations (Finished Leather)” Qualification Pack issued by “Leather Sector Skill Council”.

Sr. No.	Module	Key Learning Outcomes	Equipment Required
1	Overview on Leather Industry and Generic Skills Theory Duration (hh:mm) 08:00 Practical Duration (hh:mm) 09:00 Corresponding NOS Code Core skills/ Generic skills and Professional skills collated from all the NOS of the QP	• Explain the importance of leather industry • Describe leather manufacturing process • Explain general hazards or risk that can lead to accidents • Follow basic safety, health and hygiene measures • Wear and use PPE for safety • Carry out communication effectively with co-workers in writing as well as orally • Read the documents that are necessary for them to read to carry out operator's task	Chairs, Trainers desk and Computer, Student table, Projector, White projector screen, White board, Flip chart display stand, Safety Hazard signs, Newspaper, Stationery, PPE set
2	Assist in Wet Operations in Finished Leather Theory Duration (hh:mm) 08:00 Practical Duration (hh:mm) 81:00 Corresponding NOS Code LSS/N0901	• Identify tools and equipment used in wet operations in finished leather • Describe various process in wet operations- soaking, liming, fleshing, splitting, re-liming and de-liming • Assist in processing the material by carrying out chemical and mechanical operations • Assist in setting up of different machines used in wet operations • Prepare work area for the operations • Arrange, sort and place the leather sheets for next stage of production accordingly	De-salting machine, Un-hairing machine, Fleshing machine, Splitting machine, Sammying machine, Shaving machine, Setting machine
3	Support to Achieving Product Quality in Wet Operations Theory Duration (hh:mm) 05:00 Practical Duration (hh:mm) 33:00 Corresponding NOS Code LSS/N0902	• Explain the importance of product checks • Describe the issues pertaining to quality in leather processing • Methods of identifying faults with the finished leather	Stage wise inspection
4	Maintain the Work Area, Tools and Machines	• Ensure housekeeping and safety • Maintain workplace, tools and equipment	Spanner set, Hex key set,

Sr. No.	Module	Key Learning Outcomes	Equipment Required
	Theory Duration (hh:mm) 06:00 Practical Duration (hh:mm) 26:00 Corresponding NOS Code LSS/N8501		Screwdriver set, Cutting plier, Leather thickness gauge, Cleaning machine, Vacuum cleaner
5	Maintain Health, Safety and Security at Workplace Theory Duration (hh:mm) 5:00 Practical Duration (hh:mm) 11:00 Corresponding NOS Code LSS/N8601	- Identify and correct malfunctions in machinery and equipment - Minimize health and safety risks to self and others due to own actions - Monitor the workplace and work processes for potential risks and threats - Carry out periodic walk- through to keep work area free from hazards and obstructions, if assigned - Undertake first aid, fire-fighting and emergency response training, if asked to do so - Take action based on instructions in the event of fire, emergencies or accidents - Follow organisation procedures for shutdown and evacuation when required	First aid kit, Fire extinguisher, Earplug, Head cover, Helmet, Apron, Gumboot, Respiratory mask, Goggles, Hand gloves, Phenyl, Floor cleaning machine, Safety shower, Safety eyewash
6	Comply with Industry, Regulatory and Organizational Requirements Theory Duration (hh:mm) 03:00 Practical Duration (hh:mm) 05:00 Corresponding NOS Code LSS/N8701	- Carry out work functions in accordance with legislation and regulations, organizational guidelines and procedures - Apply and follow policies and procedures within the work practices - Identify and report any possible deviation to ethical requirements	Sample legal documents according to factory act
	Total Duration Theory Duration 35:00 Practical Duration 165:00	Unique Equipment Required: De-salting machine, Un-hairing machine, Fleshing machine, Splitting machine, Sammying machine, Shaving machine, Setting machine, Stage wise inspection, Spanner set, Hex key set, Screwdriver set, Cutting plier, Leather thickness gauge, Cleaning machine, Vacuum cleaner, First aid kit, Fire extinguisher, Earplug, Head cover, Helmet, Apron, Gumboot, Respiratory mask, Goggles, Hand gloves, Phenyl, Floor cleaning machine, Safety shower, Safety eyewash, Sample legal	

Sr. No.	Module	Key Learning Outcomes	Equipment Required
		documents according to factory act	

Grand Total Course Duration: 200 Hours, 0 Minutes

(This syllabus/ curriculum has been approved by Leather Sector Skill Council)

Trainer Prerequisites for Job role: “Helper- Wet Operations (Finished Leather)” mapped to Qualification Pack: “LSS/Q0901, Version 1.0”

Sr. No.	Area	Details
1	Description	To deliver accredited training service, mapping to the curriculum detailed above, in accordance with the Qualification Pack “LSS/Q0901, Version 1.0”.
2	Personal Attributes	Aptitude for conducting training, and pre/ post work to ensure competent, employable candidates at the end of the training. Strong communication skills, interpersonal skills, ability to work as part of a team; a passion for quality and for developing others; well-organised and focused, eager to learn and keep oneself updated with the latest in the mentioned field.
3	Minimum Educational Qualifications	10 th with 18 months certificate course in Finished Leather (any) and above.
4a	Domain Certification	Certified for Job Role: “Helper-Wet Operations (Finished Leather)”, mapped to QP: “LSS/Q0901, Version 1.0”. The minimum accepted score should be as per the decision of the respective SSC guidelines.
4b	Platform Certification	Certified for Job Role: “Trainer”, mapped to QP: “SSC/Q1402”. The minimum accepted score should be as per the decision of the respective SSC guidelines.
5	Experience	Minimum 6 years site experience with Wet Operations (Finished Leather) Level-4 qualified.

Annexure: Assessment Criteria

Assessment Criteria	
Job Role	Helper-Wet Operations (Finished Leather)
Qualification Pack	LSS/Q0901, V1.0
Sector Skill Council	Leather

Sr. No.	Guidelines for Assessment
1	Criteria for assessment for each qualification pack will be created by the sector skill council. Each performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2	The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3	Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4	Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training centre based on these criteria.
5	To pass the Qualification Pack, every trainee should score a minimum of 50% aggregate.
6	In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

Assessable Outcome	Assessment Criteria	Total Mark	Out Of	Marks Allocation	
				Theory	Skills Practical
1. LSS/N0901: Assist in wet operations in finished leather	PC1. Make sure that the work area is free from hazards	100	2	0	2
	PC2. Obtain and check equipment, protective clothing and other necessary requirements for serviceability		3	0	3
	PC3. Assist in processing the material by carrying out chemical and mechanical operations		15	10	5
	PC4. Follow the correct sequence of operations		5	0	5
	PC5. Accurately follow the essential information contained in the work ticket		3	0	3
	PC6. Handle and move the material in the most effective manner and in the correct sequence		2	0	2
	PC7. Check that the material is free from fault and fit for processing at the next stage		3	0	3
	PC8. Report any damaged work to the supervisor/quality controller		2	0	2
	PC9. Sort and place work to assist with the next stage of production and minimize the risk of damage		1	0	1
	PC10. Ask for help and information from your colleagues, supervisor when necessary, in a polite manner		2	0	2
	PC11. Anticipate and respond to requests for assistance from colleagues willingly and politely		1	0	1
	PC12. Carry the hides to and fro in the raw store for each of the preservation processes		3	0	3
	PC13. Assist in assorting the hides size wise, selection wise, grade wise and gender wise		1	0	1
	PC14. Assist in checking the hides for any defects		1	0	1
	PC15. De-salt the hides by placing the hides on the rollers of the de-salting machine		13	10	3
	PC16. Trim the unwanted edges of the hides such as legs, tail, face, udders etc		1	0	1
	PC17. Assist in the rib opening by stretching the hide		1	0	1
	PC18. Dispose the waste as per instructions		2	0	2

Assessable Outcome	Assessment Criteria	Total Mark	Out Of	Marks Allocation	
				Theory	Skills Practical
	PC19. Place hides in the soaking pit which contains water and sodium sulphide or caustic soda.		1	0	1
	PC20. Remove hides from the soaking pit after the completion of the soaking process		1	0	1
	PC21. Check if hides are clean after soaking and suitable to be passed on the liming process		1	0	1
	PC22. Carry the hides in trolleys and move it to the liming area		7	5	2
	PC23. Dispose waste as per the instructions		1	0	1
	PC24. Place hides in the liming paddle and remove the same after the stipulated period of time and move it to the fleshing area		3	0	3
	PC25. Place hides in the fleshing/ un-hairing/ scudding machine, remove the same and stretch the leather for manually scraping off with un-hairing knife any remaining organic material (connective tissue, fat etc.)		8	5	3
	PC26. Pass the unhaired hide to the splitting operation		3	0	3
	PC27. Position the hide on the bed of the splitting machine and pull it out of the machine after the operation, stacking it and pass it on for re-liming		2	0	2
	PC28. Place hides in the re-liming machine and ensure that it is removed after the inner side of the flesh has been eliminated		2	0	2
	PC29. Place and remove hides from the de-liming machine after the stipulated period of time		1	0	1
	PC30. Dispose waste as per the instructions		1	0	1
	PC31. Complete loading and batching of hides to be placed in the drums by counting, weight, volume or other defined method.		1	0	1
	PC32. Carry the hides in the trolley up to the drums for processes		2	0	2
	PC33. Remove the hides from drums and unload on to the trolley and stack it accordingly for the next stage		3	0	3

Assessable Outcome	Assessment Criteria	Total Mark	Out Of	Marks Allocation	
				Theory	Skills Practical
	of production.				
	PC34. Ensure that the drums are cleaned and ready for the various drum operations		1	0	1
	PC35. Dispose waste as per the instructions		1	0	1
	Total		100	30	70
2. LSS/N0902: Contribute to achieving product quality in wet operations	PC1. Assist in ensuring that the hides are graded and sorted as per their weight, type, and sex, and then properly trimmed	50	1	0	1
	PC2. Assist in ensuring the soaking is done properly and hide is completely clean which is important for optimum tanning to take place		6	5	1
	PC3. Ensure that excess flesh and fatty tissue from under the pelt is removed completely		11	10	1
	PC4. Ensure that de-fleshed pelt is trimmed i.e., sections of the root of the tail, the navel, the kneecaps, and other unwanted parts of the hide are cut off		1	0	1
	PC5. Ensure that the hide/crust leather is of required thickness using the gauge meter before and after splitting		1	0	1
	PC6. Assist in ensuring that regular checks will be made to see the penetration of the tanning agents, by cutting the cross section of the pelt, and observing the degree of penetration		11	10	1
	PC7. Assist in ensuring that the required part of water is pressed out after the sammying process and it is cut into sides for further processing		1	0	1
	PC8. Assist in ensuring product quality by maintaining: - Ph value - Oil and fat standard - Chemical standard		1	0	1
	PC 9. Assist in applying quality requirements for the hides for all stages of the process		6	5	1
	PC 10. Assist in conducting visual inspection continuously to ensure standard is maintained		1	0	1

Assessable Outcome	Assessment Criteria	Total Mark	Out Of	Marks Allocation	
				Theory	Skills Practical
	PC11. Recognize, isolate and report any deviations from normal		1	0	1
	PC12. Identify faults and irregularities in equipment and machinery and take action within the limits of your responsibility		6	5	1
	PC13. Maintain the continuity of production with minimum interruptions and downtime		1	0	1
	PC14. Follow reporting procedures		1	0	1
	PC15. Give proper inputs to concerned person to enable maintenance of records and documentation		1	0	1
	Total		50	35	15
3. LSS/N8501: Maintain the work area, tools and machines	PC1. Handle materials, machinery, equipment and tools safely and correctly	50	3	0	3
	PC2. Use correct lifting and handling procedures		3	2	1
	PC3. Use materials to minimize waste		2	0	2
	PC4. Prepare and organize work		2	0	2
	PC5. Maintain a clean and hazard free working area		3	2	1
	PC6. Deal with work interruptions		1	0	1
	PC7. Move around the workplace with care		2	0	2
	PC8. Maintain tools and equipment		2	0	2
	PC9. Carry out running maintenance within agreed schedules		3	1	2
	PC10. Carry out maintenance and/or cleaning outside responsibility		2	0	2
	PC11. Report unsafe equipment and other dangerous occurrences		2	0	2
	PC12. Ensure that the correct machine guards are in place		4	2	2
	PC13. Work in a comfortable position with the correct posture		3	0	3
	PC14. Use cleaning equipment and methods appropriate for the work to be carried out		2	1	1
	PC15. Dispose of waste safely in the designated location		2	1	1
	PC16. Store cleaning equipment safely after use		2	1	1
	PC17. Complete and store accurate records and documentation		2	1	1

Assessable Outcome	Assessment Criteria	Total Mark	Out Of	Marks Allocation	
				Theory	Skills Practical
	PC18. Maintain proper lighting, ventilation to make sure general comfort is there while working		2	1	1
	PC19. Give inputs and assist in completing documentation		2	1	1
	PC20. Report the need for maintenance and/or cleaning outside your area of responsibility		2	1	1
	PC21. Ensure safe and correct handling of materials, equipment and tools		2	1	1
	PC22. Maintain appropriate environment to protect stock from pilfering, theft, damage and deterioration		2	0	2
	Total		50	15	35
4. LSS/N8601: Maintain health, safety and security at workplace	PC1. Comply with health and safety related instructions applicable to the workplace	25	1	0	1
	PC2. Use and maintain personal protective equipment as per protocol		0.5	0	0.5
	PC3. Carry out own activities in line with approved guidelines and procedures		1	0	1
	PC4. Maintain a healthy lifestyle and guard against dependency on intoxicants		11	10	1
	PC5. Follow environment management system related procedures		0.5	0	0.5
	PC6. Identify and correct (if possible) malfunctions in machinery and equipment		0.5	0	0.5
	PC7. Report any service malfunctions that cannot be rectified		1	0	1
	PC8. Store materials and equipment in line with manufacturer's and organizational requirements		1	0	1
	PC9. Safely handle and move waste and debris		1	0	1
	PC10. Minimize health and safety risks to self and others due to own actions		1	0	1
	PC11. Seek clarifications, from supervisors or other authorized personnel in case of perceived risks		0.5	0	0.5
	PC12. Monitor the workplace and work processes for potential risks and threats		1	0	1

Assessable Outcome	Assessment Criteria	Total Mark	Out Of	Marks Allocation	
				Theory	Skills Practical
	PC13. Carry out periodic walk-through to keep work area free from hazards and obstructions, if assigned		0.5	0	0.5
	PC14. Report hazards and potential risks/ threats to supervisors or other authorized personnel		1	0	1
	PC15. Participate in mock drills/ evacuation procedures organized at the workplace		1	0	1
	PC16. Undertake first aid, fire-fighting and emergency response training, if asked to do so		1	0	1
	PC17. Take action based on instructions in the event of fire, emergencies or accidents		0.5	0	0.5
	PC18. Follow organization procedures for shutdown and evacuation when required		1	0	1
	Total		25	10	15
5. LSS/N8701: Comply with industry, regulatory and organizational requirements	PC1. Carry out work functions in accordance with legislation and regulations, organizational guidelines and procedures	25	9	5	4
	PC2. Seek and obtain clarifications on policies and procedures, from the supervisor or other authorized personnel		4	0	4
	PC3. Apply and follow these policies and procedures within the work practices		7	5	2
	PC4. Provide support to the supervisor and team members in enforcing these considerations		3	0	3
	PC5. Identify and report any possible deviation to these requirements		2	0	2
	Total		25	10	15
	Grand Total	250	250	100	150